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STD::SIMD IS A RANGE

ABSTRACT

P1928 “std::simd – merge data-parallel types from the Parallelism TS 2” promised a paper on making `simd` a range. This paper explores the addition of iterators to `basic_simd` and `basic_simd_mask`.

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1

CHANGELOG

1.1

CHANGES FROM REVISION 0

Previous revision: P3480R0

- Simplify to a single iterator class template.
- Remove incorrect `operator-` overload.
- Discuss design choice of using a sentinel type for `end()`.

1.2

CHANGES FROM REVISION 1

Previous revision: P3480R1

- Add SG9 poll results.
- Use `default_sentinel_t` instead of a new sentinel type.
- Use an almost-mutable iterator type as directed by SG9 for non-const `begin()`.
- Fix `for_each` example to use `ranges` version.

1.3

CHANGES FROM REVISION 2

Previous revision: P3480R2

- Ask about tuple interface for `simd`.
- Clarify where `[simd.iterator]` should go.
- Provide proper wording.
- Fix `<=>` comparison with `default_sentinel_t`.
- Bump feature test macro?

2

STRAW POLLS

2.1

SG9 AT WROCŁAW 2024

Poll: We want `std::basic_simd` to be a range.

SF	F	N	A	SA
6	2	0	0	0

Poll: We want `std::basic_simd` to be a common range.

SF	F	N	A	SA
0	0	3	4	1

Poll: We want `std::basic_simd::operator[]` and `std::basic_simd::begin/end` in C++26 without mutation support, knowing that we might not be able to do it later due to ABI issues (e.g. `decltype(auto) f(std::simd<float> x) { return x[0]; }` could change return type).

SF	F	N	A	SA
6	2	0	0	0

Poll: We want `std::basic_simd::iterator` and `std::basic_simd::const_iterator` to be different types to make the transition to mutable iteration easier. This also means adding a non-const `begin()` overload that returns a different type than the const `begin()` overload but currently has the same semantics.

SF	F	N	A	SA
2	3	3	0	0

Poll: Use `std::default_sentinel_t` instead of *simd-iterator-sentinel*.

→ unanimous consent

Poll: Forward P3480R1 with the changes above to LEWG for inclusion in C++26.

SF	F	N	A	SA
7	1	0	0	0

3

INTRODUCTION, OR WHY SIMD WASN'T A RANGE IN THE TS

The Parallelism TS 2 was based on C++17. Ranges were added in C++20. Before ranges, an iterator category was tied to whether `operator*` of iterators returned an lvalue reference. Since `basic_simd` and `basic_simd_mask` objects are not composed of sub-objects (in other words, a `simd<int>` contains no *int objects*), `operator[]` returns prvalues (or a proxy reference in the TS for the non-const case). An iterator needs to do the same and thus never could be in any other iterator category than *Cpp17InputIterator*. In reality, the iterator category always was “random access” (never contiguous; because while `basic_simd` is a contiguous range in memory it isn't one in the object model of C++). In order to not cement that mismatch, it was never proposed to make `basic_simd/basic_simd_mask` a range for the TS.

Now that the iterator concepts don't require an lvalue reference anymore we can easily make `basic_simd/basic_simd_mask` a read-only range. Iterator dereference would return a prvalue (a copy of the value stored in the `basic_simd/basic_simd_mask` object). In addition, the abstraction of a

sentinel instead of an iterator pointing beyond the last value of the `basic_simd` seems like a useful tool for `basic_simd`.

4

MOTIVATION

After the technical reasons for *not* adding iterators to `basic_simd/basic_simd_mask` are resolved, we still need to consider why `basic_simd` should be a range in the first place.

5

INTEGRATION WITH THE STANDARD LIBRARY

We can improve integration of `basic_simd/basic_simd_mask` with the rest of the standard library. By making `basic_simd/basic_simd_mask` a range many of the existing facilities in the standard library become easily accessible. All of these facilities do work as intended — in other words: presenting `basic_simd/basic_simd_mask` as a range matches on the semantic level, not only syntactically.

5.1

READ-ONLY SUBSCRIPT SHOULD IMPLY READ-ONLY ITERATION

With P1928R12 we can write

```
std::simd<int> v = ...;
for (int i = 0; i < v.size(); ++i) {
    do_something(v[i]);
}
```

Why then, can we not also write

```
for (auto x : v) {
    do_something(x);
}
```

and

```
std::ranges::for_each(v.begin(), v.end(), [](auto x) {
    do_something(x);
});
```

and

```
v | std::views::filter([](auto x) { return x > 0; }) | std::ranges::to<std::vector>();
```

C++ users have learned that whenever a for loop with subscript does what they need to do, then a ranged for loop, standard algorithm, or range adaptor are valid alternatives. This expectation should not get an exception with `basic_simd` and `basic_simd_mask`.

5.2

PRESENT A RANGE OF SIMD AS A RANGE OF SIMD'S VALUE-TYPE

In some applications it is more efficient (and simpler) to work with `basic_simd` objects internally, instead of constantly doing loads and stores. Thus a fairly simple container that comes up in applications could be `std::vector<std::simd<float>>`. On I/O such an application typically cannot communicate in `basic_simd` objects anymore. Instead it needs to present a range of `float`s. Read-only iterators on `basic_simd` do not help with the input side. But for output we can easily turn the `vector<simd<float>>` into a range of `float`:

```
std::vector<std::simd<float>> data;
auto range_of_float = data | std::views::join;
```

6

DOWNSIDES OF MAKING SIMD A RANGE

Really, I can't think of any downsides of making `basic_simd/basic_simd_mask` a range. In principle one could argue that `basic_simd/basic_simd_mask` is not a container [P0851R0]. Consequently, it shouldn't have a container interface and thus no iterators. But then we should probably remove the subscript operator as well.

7

DESIGN CHOICE: SENTINEL

The `basic_simd` iterator type must have a reference/pointer to the `basic_simd` object it is iterating together with an offset, where into the `basic_simd` it is pointing. Because of these two members (and their type), the iterator already knows the complete bounds of the range it is pointing into. Consequently, a single `basic_simd` iterator can always determine whether it points at the beginning or end of the range, it doesn't need to compare against another offset. A sentinel type allows asking that question via `operator==`. Thus, instead of comparing two runtime offset members on `operator==`, a compare against a sentinel is implemented as a compare against a compile-time constant. This makes it easier for the compiler to optimize and reduces the size of the `end()` sentinel to a single byte (empty type).

8

OPEN QUESTIONS

8.1

MAKE ITERATOR CONVERTIBLE TO CONST_ITERATOR

After SG9 voted to make `basic_simd::const_iterator` and `basic_simd::iterator` different types, interaction between the two types needs to be considered. Since iterators model pointers and non-const pointers convert to const pointers, it seems we need conversion from `simd-iterator<V>` to `simd-iterator<const V>` to be implicit. In the wording below, “`#define LEWG_WANTS_CONVERSION 1`” to make `v.begin() == v.cbegin()` a valid expression (evaluating to `true`).

8.2

ADD TUPLE INTERFACE

`std::array` implements the tuple interface. Should `std::simd` also implement `tuple_size`, `tuple_element`, and `get`?

9

WORDING

9.1

FEATURE TEST MACRO

In [version.syn] bump the `__cpp_lib_simd` version.

9.2

ADD [SIMD.ITERATOR]

Add a new subclause before §29.10.6 [simd.class]:

[simd]

(9.2.1) 29.10.6 Class *simd-iterator*

[simd.iterator]

```

namespace std {
    template <typename V>
    class simd-iterator {           // exposition only
        V* data_ = nullptr;        // exposition only
        simd-size-type offset_ = 0; // exposition only

    public:
        using value_type = typename V::value_type;
        using iterator_category = std::random_access_iterator_tag;
        using difference_type = simd-size-type;

        constexpr simd-iterator() = default;
        constexpr simd-iterator(V& d, int off);

        constexpr simd-iterator(const simd-iterator&) = default;
        constexpr simd-iterator& operator=(const simd-iterator&) = default;

    #if LEWG_WANTS_CONVERSION
        constexpr simd-iterator(const simd-iterator<remove_const_t<V>>&) requires is_const_v<V>;
    #endif

        constexpr value_type operator*() const;

        constexpr simd-iterator& operator++();
        constexpr simd-iterator operator++(int);
        constexpr simd-iterator& operator--();
        constexpr simd-iterator operator--(int);

```

```

constexpr simd-iterator& operator+=(difference_type n);
constexpr simd-iterator& operator-=(difference_type n);

constexpr value_type operator[](difference_type n) const;

constexpr friend bool operator==(simd-iterator a, simd-iterator b) = default;
constexpr friend bool operator==(simd-iterator a, default_sentinel_t);
constexpr friend auto operator<=>(simd-iterator a, simd-iterator b);
constexpr friend auto operator<=>(simd-iterator a, default_sentinel_t);

constexpr friend simd-iterator operator+(const simd-iterator& i, difference_type n);
constexpr friend simd-iterator operator+(difference_type n, const simd-iterator& i);
constexpr friend simd-iterator operator-(const simd-iterator& i, difference_type n);

constexpr friend difference_type operator-(simd-iterator a, simd-iterator b);
constexpr friend difference_type operator-(simd-iterator i, default_sentinel_t);
constexpr friend difference_type operator-(default_sentinel_t, simd-iterator i);
};
}

```

```
constexpr simd-iterator(V& d, int off);
```

1 *Effects:* Initializes *data_* with *d* and *offset_* with *off*.

```
#if LEWG_WANTS_CONVERSION
```

```
constexpr simd-iterator(const simd-iterator<remove_const_t<V>>& i) requires is_const_v<V>;
#endif
```

2 *Effects:* Initializes *data_* with *i.data_* and *offset_* with *i.offset_*.

```
constexpr value_type operator*() const;
```

3 *Preconditions:*

- *data_* is a valid pointer, and
- *offset_* is in the range $[0, V::size())$.

Fixme: *Preconditions:* already implied by *Effects:* ?

4 *Effects:* Equivalent to: `return (*data_)[offset_];`

```
constexpr simd-iterator& operator++();
```

5 *Effects:* Equivalent to:

```

++offset_;
return *this;

```

```
constexpr simd-iterator operator++(int);
```

6 *Effects:* Equivalent to:
 simd-iterator tmp = *this;
 ++*offset_*;
 return tmp;

```
constexpr simd-iterator& operator--();
```

7 *Effects:* Equivalent to:
 --*offset_*;
 return *this;

```
constexpr simd-iterator operator--(int);
```

8 *Effects:* Equivalent to:
 simd-iterator tmp = *this;
 --*offset_*;
 return tmp;

```
constexpr simd-iterator& operator+=(difference_type n);
```

9 *Effects:* Equivalent to:
 offset_ += n;
 return *this;

```
constexpr simd-iterator& operator-=(difference_type n);
```

10 *Effects:* Equivalent to:
 offset_ -= n;
 return *this;

```
constexpr value_type operator[](difference_type n) const;
```

11 *Effects:* Equivalent to: return (**data_*)[*offset_* + n];

```
constexpr friend bool operator==(simd-iterator i, default_sentinel_t);
```

12 *Effects:* Equivalent to: return i.*offset_* == V::size();

```
constexpr friend auto operator<=>(simd-iterator a, simd-iterator b);
```

13 *Preconditions:* `a.data_ == b.data_` is true.

14 *Effects:* Equivalent to: `return a.offset_ <=> b.offset_;`

`constexpr friend auto operator<=>(simd-iterator i, default_sentinel_t);`

15 *Effects:* Equivalent to: `return i.offset_ <=> V::size();`

`constexpr friend simd-iterator operator+(const simd-iterator& i, difference_type n);`

`constexpr friend simd-iterator operator+(difference_type n, const simd-iterator& i);`

16 *Effects:* Equivalent to: `return simd-iterator(*i.data_, i.offset_ + x);`

`constexpr friend simd-iterator operator-(const simd-iterator& i, difference_type n);`

17 *Effects:* Equivalent to: `return simd-iterator(*i.data_, i.offset_ - x);`

`constexpr friend difference_type operator-(simd-iterator a, simd-iterator b);`

18 *Preconditions:* `a.data_ == b.data_` is true.

19 *Effects:* Equivalent to: `return a.offset_ - b.offset_;`

`constexpr friend difference_type operator-(simd-iterator i, default_sentinel_t);`

20 *Effects:* Equivalent to: `return i.offset_ - V::size();`

`constexpr friend difference_type operator-(default_sentinel_t, simd-iterator i);`

21 *Effects:* Equivalent to: `return V::size() - i.offset_;`

9.3

MODIFY [SIMD.OVERVIEW]

[simd.overview]

```

template<class T, class Abi> class basic_simd {
public:
    using value_type = T;
    using mask_type = basic_simd_mask<sizeof(T), Abi>;
    using abi_type = Abi;
    using iterator = simd-iterator<basic_simd>;
    using const_iterator = simd-iterator<const basic_simd>;

    constexpr iterator begin();
    constexpr const_iterator begin() const;
    constexpr const_iterator cbegin() const;
    constexpr default_sentinel_t end() const;
    constexpr default_sentinel_t cend() const;

```

9.4

MODIFY [SIMD.MASK.OVERVIEW]

[simd.mask.overview]

```

template<size_t Bytes, class Abi> class basic_simd_mask {
public:
    using value_type = bool;
    using abi_type = Abi;
    using iterator = simd-iterator<basic_simd_mask>;
    using const_iterator = simd-iterator<const basic_simd_mask>;

    constexpr iterator begin();
    constexpr const_iterator begin() const;
    constexpr const_iterator cbegin() const;
    constexpr default_sentinel_t end() const;
    constexpr default_sentinel_t cend() const;

```

[simd.tuple]

A

BIBLIOGRAPHY

- [P0851R0] Matthias Kretz. *P0851R0: `simd<T>` is neither a product type nor a container type*. ISO/IEC C++ Standards Committee Paper. 2017. URL: <https://wg21.link/p0851r0>.